

1 Write down the steps involved in PCM for baseband signal.

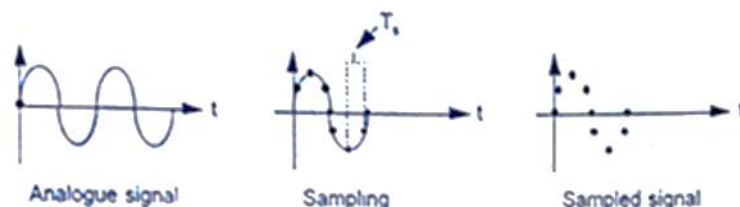
BUET M.Sc. 12

Solution: PCM involves three main steps:

- Sampling
- Quantization
- Coding

Sampling

Sampling involves measuring the analog signal at specific time intervals.



The accuracy of describing the analog signal in digital terms depends on how often the analog signal is sampled. This is expressed as the sampling frequency. The sampling theory states that:

To reproduce an analog signal without distortion, the signal must be sampled with at least twice the frequency of the highest frequency component in the analog signal.

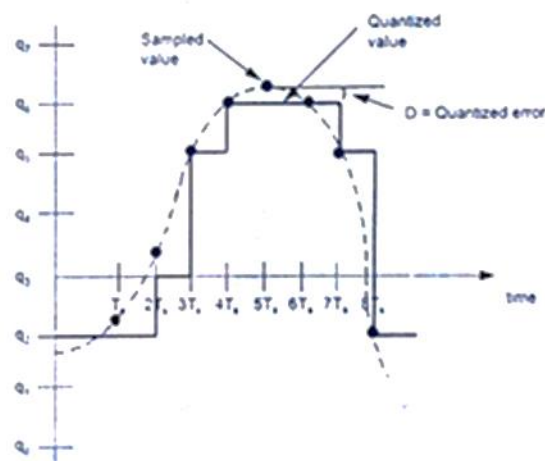


Figure: Quantization

2 State Nyquist theorem. What is the condition to recover message signal from sampled signal.

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M.Sc. 13

Solution: Nyquist theorem is a theorem that is followed in the digitization of analog signals. It states that-

"A signal must be sampled at least twice as fast as the bandwidth of the signal to accurately reconstruct the waveform; otherwise, the high-frequency content will alias at a frequency inside the spectrum of interest (passband)."

To reproduce an analog signal without distortion, the signal must be sampled with at least twice the frequency of the highest frequency component in the analog signal.

12 Calculate the bit rate of a single channel PCM system that uses the standard band limited telephone channel.

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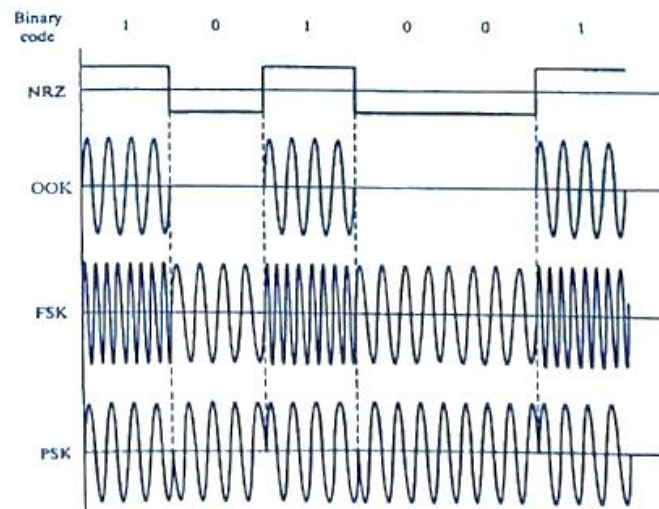
Solution:

$$f_s = 8 \text{ kHz} \quad (\text{standard sampling frequency})$$

$$n = 8 \quad [\text{standard no. of bit for PCM is 8. 7 bit is used for encoding and 1 bit is used to indicate sign (MSB)}]$$

$$\therefore R_b = n f_s = 8 \times 8 = 64 \text{ kbps is standard bit rate for a single channel PCM.}$$

Solution:



54

The output power from an audio amplifier is 10 W when the signal frequency is 1 kHz. At a signal frequency of 10 kHz, the output power is 1 W. Calculate the dB change in power.

- (a) +20 dB (b) +10 dB (c) -25 dB (d) -10 dB

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Explanation: Ans: D.

$$\text{dB change in power} = 10 \log \left(\frac{P_2}{P_1} \right) = 10 \log \left(\frac{1}{10} \right) = 10 \times (-1) = -10 \text{ dB}$$

60

If a 300-3300 Hz channel have 2 MW signal power and 1.95 MW noise power, then find the channel capacity.

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Solution:

$$\text{Bandwidth, } B = 3300 - 300 = 3000 \text{ Hz.}$$

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

$$= 3000 \log_2 \left(1 + \frac{2}{1.95} \right)$$

$$= 3000 \log_2 (1 + 1.02564)$$

$$= 3000 \log_2 (2.02564)$$

$$= 3000 \times 1.018378$$

$$= 3055.135 \text{ bit/s } \underline{\text{Ans.}}$$

62

A voice grade channel of telephone network has a bandwidth of 3.4 kHz. Calculate the information capacity of the channel for a signal to Noise Ratio of 30dB.

DPDC-16,
BUET M.Sc.-16

Solution:

$$C = B \log_2 (1 + \text{SNR})$$

$$= 3400 \log_2 1001$$

$$= 3400 \times 9.967$$

$$= 33887.8 \text{ bit/sec}$$

$$= 33.89 \text{ kbit/sec}$$

Ans.

$$B = 3.4 \text{ kHz} = 3400 \text{ Hz}$$

$$\text{SNR} = 30 \text{ dB}$$

$$= 10^3$$

$$= 1000$$

12	A PCM-TDM system multiplexes 10 band limited voice channel (300-3400 Hz) and uses a 256 level quantizer. If the signal is sampled at a rate $17\frac{11}{17}\%$ higher than Nyquist rate, then what will be the maximum energy bandwidth of the transmission channel?	DPDC-14
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Solution: Here,

Maximum frequency of the message signal, $f_m = 3400$ Hz.

$\therefore$ Nyquist frequency of the signal, $f_{NQ} = 2 f_m = 2 \times 3400$ Hz = 6.8 KHz

$\therefore$ Sampling frequency of the signal, $f_s = 17\frac{11}{17}\%$ (=17.65%) higher than Nyquist rate
 $= 1.1765 \times 6.8$ KHz = 8KHz

Quantization Level, $L = 256 = 2^8 = 2^n$

$\therefore$ No. of bits in the code, $n=8$

$\therefore$ Maximum energy bandwidth of one channel = $n f_s = 8 \times 8$ KHz = 64 kbit/sec

$\therefore$ Maximum energy bandwidth of 10 channel = 10×64 KHz = 640 kbit/sec **Ans.**

69	$x(t) = 1.5 \cos(800\pi t)$ is to be PCM with minimum SQNR of 25 dB. How many bytes are required for encoding each having uniform quantization.	BUET MSC-14
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Solution: We Know,

$$SNR_{dB} = 1.76 + 6.02 n$$

$$25 = 1.76 + 6.02 n$$

$$n = \frac{25-1.76}{6.02} = 3.86 \approx 4 \text{ bits} = \frac{4}{8} \text{ byte} = 0.5 \text{ byte} \quad \text{Ans.}$$

70	In a PCM system, an 8 kHz sinusoidal signal is uniformly sampled with a rate twice the nyquist frequency. If the bit rate is 160 kbps, Determine the SNR of the system in dB. (a) 25.84 dB (b) 31.86 dB (c) 43.9 dB (d) 55.94 dB	BUET M.Sc.-22 (June)
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Solution: Ans.: (b) 31.86 dB

$$f_s = 2 f_{nq} = 2 \times 2 f_m = 4 f_m = 4 \times 8 = 32 \text{ kHz}$$

$$R_B = n f_s = 160 \text{ kbps}$$

$$\therefore n = \frac{R_B}{f_s} = \frac{160}{32} = 5$$

$$SNR_{(in \text{ dB})} = 1.76 + 6.02 n = 1.76 + (6.02 \times 5) = 31.86 \text{ dB}$$

49	$x(t) = 1.5 \cos(800\pi t)$ is to be PCM with minimum SQNR of 25 dB. How many bytes are required for encoding each having uniform quantization.	BUET MSC-14
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32	Consider PCM system with a uniform quantization of step size 2 V. The input is a sinusoidal signal with amplitude of 10 V. Calculate the signal to noise quantization ratio in dB and number of bits per sample to be used in the system.	BTCL-21
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Solution:

$$\Delta = \frac{2m_p}{L}$$

$$\Rightarrow L = \frac{2m_p}{\Delta} = \frac{2 \times 10}{2} = 10$$

But no. of quantization level should be power of 2.

$$\therefore L = 16 = 2^4 = 2^n$$

$$\therefore n = 4 \quad \text{Ans.}$$

Step size, $\Delta = 2$ V

Amplitude, $m_p = 10$ V

Let, No. of quantization level = L

$SQNR_{dB} = ?$

Number of bits per sample, $n = ?$

$$SQNR_{dB} = 1.76 + 6.02 n$$

$$= 1.76 + 6.02 \times 4$$

$$= 25.84 \text{ dB} \quad \text{Ans.}$$

31	In a PCM system, an 8 kHz sinusoidal signal is uniformly sampled with a rate 75% higher than nyquist frequency. The quantizer uses a rounding operation and the maximum quantization error is 4% of the peak amplitude of the signal. Determine the bit rate and SNR of the system in dB.	BUET M.Sc.-19
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Solution:

$$f_s = 8 \times 2 \times 1.75 = 28 \text{ kHz}$$

$$\Delta/2 \leq 0.04 m_p$$

$$(2m_p/L)/2 \leq 0.04 m_p$$

$$L \geq 25$$

$$L \approx 32$$

$$n=5$$

$$R_B = n f_s = 5 \times 28 = 140 \text{ kbps}$$

$$SNR \text{ (in dB)} = 1.76 + 6.02 n = 31.86 \text{ dB}$$

33	$x(t) = 200 \sin(1000\pi t) + 500 \sin(2000\pi t)$. The signal is sampled at a rate $33\frac{1}{3}\%$ higher than the Nyquist rate. If Delta Modulation is used, find the bit rate & Baud Rate. If QPSK is used, find the information rate.	BUET M.Sc.-16
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Solution:

Maximum signal frequency, $f_m = 2000\pi / 2\pi = 1000 \text{ Hz} = 1 \text{ kHz}$.

Nyquist frequency, $f_{NQ} = 2f_m = 2 \times 1 \text{ kHz} = 2 \text{ kHz}$

Sampling frequency, $f_s = 1.3333 \times f_{NQ} = 1.3333 \times 2 \text{ kHz} = 2.6666 \text{ kHz}$

For Delta Modulation,

Quantization Level $L = 2 = 2^1 = 2^n$

No. of bits, $n = 1$

Bit Rate, $R_B = nf_s = 1 \times 2.6666 = 2.6666 \text{ kbps}$ Ans.

Baud Rate, $R_B / n = 2.6666 / 1 = 2.6666 \text{ kbps}$ Ans.

For QPSK,

Quantization Level $L = 4 = 2^2 = 2^n$

No. of bits, $n = 2$

Bit Rate, $R_B = nf_s = 2 \times 2.6666 = 5.3332 \text{ kbps}$ Ans.

Baud Rate, $R_B / n = 5.3332 / 2 = 2.6666 \text{ kbps}$ Ans.

[Note: $33\frac{1}{3}\%$ ($= 0.3333$) times higher $= 1.3333$ times]

98	If $m(t) = B \text{ Sinc}(2\pi Bt)$, $B=1000 \text{ Hz}$ & $\omega_c = 10000\pi$, then Draw the Spectrum of DSB-SC and LSB signal.	BUET M. - 12
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Solution:

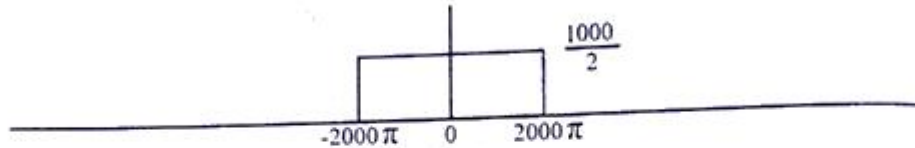


Figure: $M(\omega)$

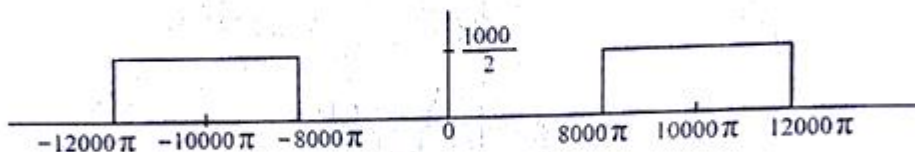


Figure: DSB-SC

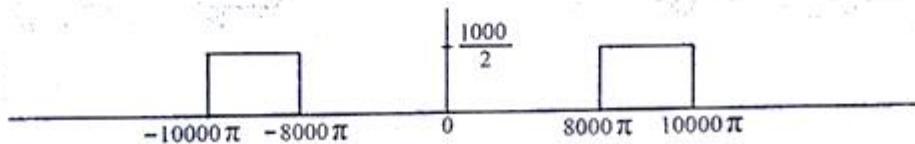
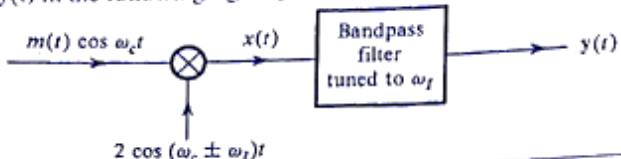


Figure: LSB

121	Similar: Find $y(t)$ in the following figure [Example 4.2, B.P. Lathi] 	BUET M.Sc.-19
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Solution:

Multiplication with $2 \cos (\omega_c \pm \omega_1)t$ yields,

$$\begin{aligned}
 x(t) &= m(t) \cos \omega_c t \times 2 \cos (\omega_c \pm \omega_1)t \\
 &= m(t) \times [2 \cos \omega_c t \cos (\omega_c \pm \omega_1)t] \\
 &= m(t) \times [\cos \{\omega_c - (\omega_c \pm \omega_1)\}t + \cos \{\omega_c + (\omega_c \pm \omega_1)\}t] \\
 &= m(t) \times [\cos (\pm \omega_1)t + \cos (2\omega_c \pm \omega_1)t] \\
 &= m(t) \cos (\pm \omega_1)t + m(t) \cos (2\omega_c \pm \omega_1)t
 \end{aligned}$$

As band pass filter is tuned to ω_1 frequency it will only allow signals having ω_1 frequency. So, signals having other frequencies $[-\omega_1 \& (2\omega_c \pm \omega_1)]$ will be filtered out.

$$\therefore y(t) = m(t) \cos \omega_1 t \quad \text{Ans.}$$

134	$x(n) = \sin (25 \pi n)$. Find the period of $x(n)$. Considering the standard sampling rate for telephone system find the bit rate if Delta Modulation is used.	BUET M.Sc.-17
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Solution:

$$x(n) = \sin (25 \pi n) = \sin \left\{ 2\pi \left(\frac{25}{2} \right) n \right\}$$

$$\text{So, frequency } f = \frac{25}{2} \text{ Hz} = 12.5 \text{ Hz}$$

$$\text{Time period, } T = \frac{1}{f} = \frac{1}{12.5} = 0.08 \text{ second.} \quad \text{Ans.}$$

Standard sampling rate for telephone system, $f_s = 8 \text{ kHz}$

For, delta modulation, no. of bit $n = 1$

$$\text{So, Bit rate } R_B = n f_s = 1 \times 8 \text{ kHz} = 8 \text{ kHz} \quad \text{Ans.}$$

135	What is erlang of telephone traffic? Related Math.	BUET M.Sc. Unknown
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Solution:

Erlang is a unit of telecommunications traffic measurement. It is used to describe the total traffic volume of one hour.

For example, if a group of user made 30 calls in one hour, and each call had an average call duration of 5 minutes, then the number of Erlangs this represents is worked out as follows:

Minutes of traffic in the hour = number of calls $\times$ duration

$$= 30 \times 5 = 150$$

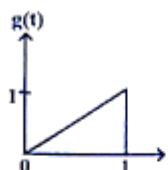
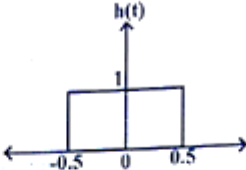
Hours of traffic in the hour = $150 / 60$

$$= 2.5$$

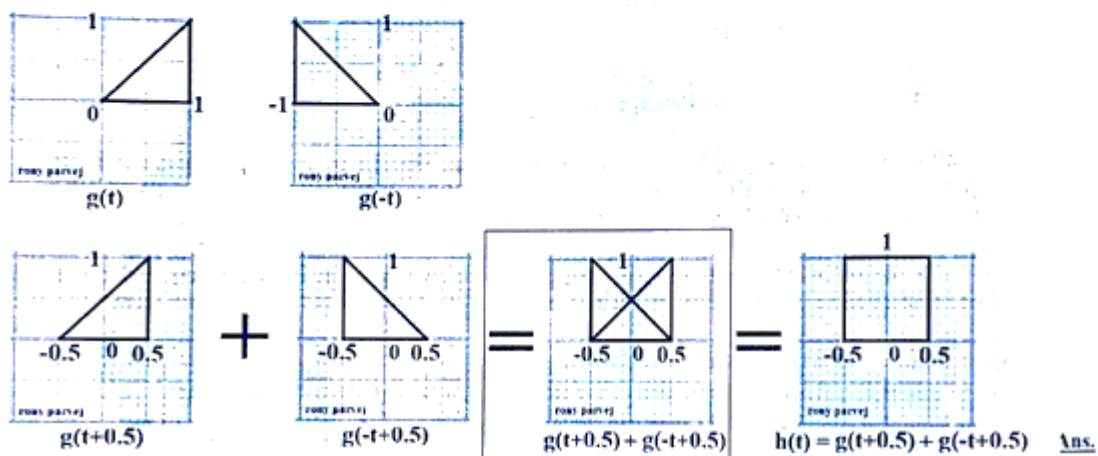
Traffic figure = 2.5 Erlangs

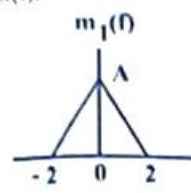
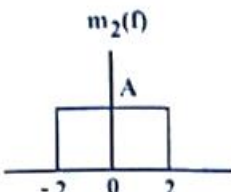
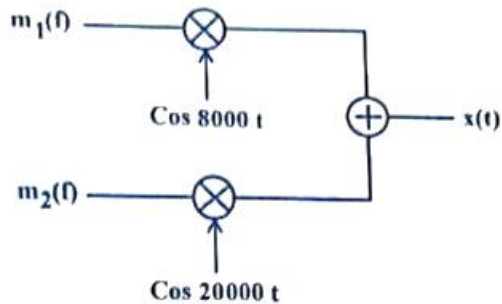
Erlang traffic measurements are made in order to help telecommunications network designers understand traffic patterns within their voice networks. This is essential if they are to successfully design their network topology and establish the necessary trunk group sizes.

Erlang traffic measurements or estimates can be used to work out how many lines are required between a telephone system and a central office (PSTN exchange lines), or between multiple network locations.

137	Express $h(t)$ as a combination of $g(t)$ using scaling, shifting or mirroring if necessary.  $g(t)$  $h(t)$	BUET M.Sc.-18
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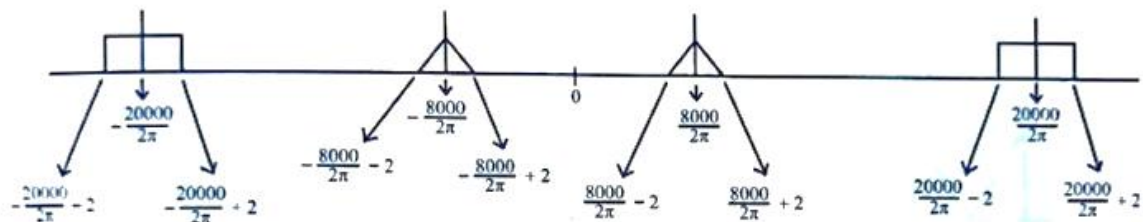
Solution:



65	Find $x(t)$.  $m_1(f)$  $m_2(f)$ 	BUET M.Sc.-17
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Solution:

Solution:



135	What is erlang of telephone traffic? Related Math.	BUET M.Sc. Unknown
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For example, if a group of user made 30 calls in one hour, and each call had an average call duration of 5 minutes, then the number of Erlangs this represents is worked out as follows:

$$\begin{aligned}
 \text{Minutes of traffic in the hour} &= \text{number of calls} \times \text{duration} \\
 &= 30 \times 5 = 150 \\
 \text{Hours of traffic in the hour} &= 150 / 60 \\
 &= 2.5 \\
 \text{Traffic figure} &= \mathbf{2.5 \text{ Erlangs}}
 \end{aligned}$$

Erlang traffic measurements are made in order to help telecommunications network designers understand traffic patterns within their voice networks. This is essential if they are to successfully design their network topology and establish the necessary trunk group sizes.

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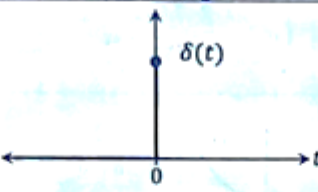
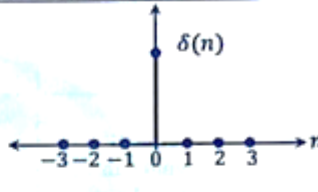
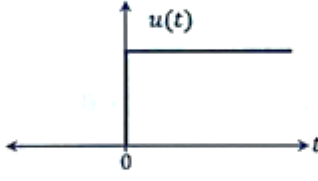
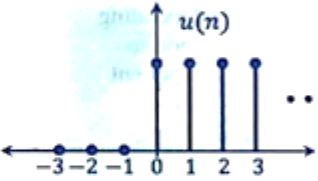
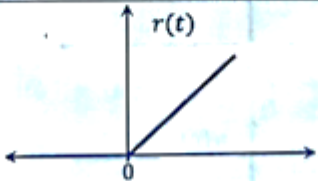
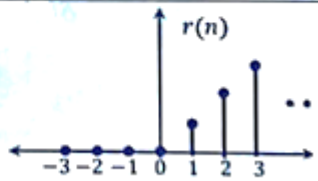
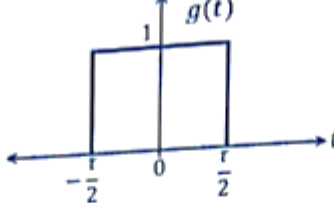
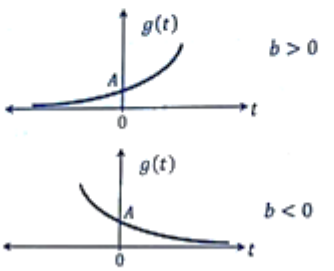
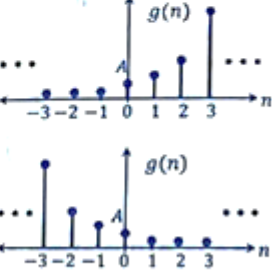
136	Suppose a call-center has 25 phone lines, 1000 calls per day and the average duration of a call is 15 minutes. Calculate offered load of the call center in Erlang. Also Calculate the maximum number of calls the call center can serve per day.	ERL-17
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Solution:

$$\text{Offered load} = \frac{\text{Minutes of traffic in a day}}{\text{Minutes in a day}} = \frac{1000 \times 15}{60 \times 24} = 10.42 \text{ Erlang}$$

$$\text{Maximum number of calls the call center can serve per day} = 10.42 \times 24 = 250.08 \text{ Erlang} \quad \mathbf{Ans.}$$

Elementary Signals:

Name	Definition	Continuous Time Signal	Discrete Time Signal
Unit Impulse Signal	An impulse signal has zero value except at $t = 0$. It has infinitely high value at $t = 0$.	 $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$	 $\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$
Unit Step Signal	A unit step signal has unity value for $t \geq 0$ else zero value.	 $u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$	 $u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$
Unit Ramp Signal	A ramp step signal has unity slope value for $t \geq 0$, otherwise it has zero value.	 $r(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$	 $r(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$
Rectangular Pulse Signal Or, Gate pulse, Or, Pulse function, or Window function	A unit rectangular pulse has unit amplitude within a time interval, otherwise it has zero value.	 $g(t) = \begin{cases} 1 & -\frac{\tau}{2} \leq t \leq +\frac{\tau}{2} \\ 0 & \text{Otherwise} \end{cases}$	 $g(n) = \begin{cases} 1 & -m \leq n \leq +m \\ 0 & \text{Otherwise} \end{cases}$
Exponential Signal	An exponential signal can either have exponentially rising or falling amplitude depending upon its exponent value.	 $g(t) = Ae^{bt} \quad A > 0$	 $g(n) = Ae^{bn} \quad A > 0$

Comparison of Multiple Access Techniques

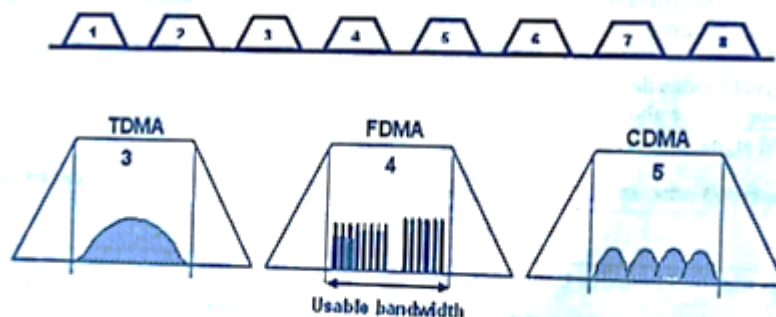
Feature	FDMA	TDMA	CDMA
Division Method	Frequency bands	Time slots	Spreading codes
Technology	Analog/Digital	Digital	Digital
Simultaneous Users	Same time, different frequencies	Same frequency, different times	Same time and frequency
Capacity	Limited by frequency bands	Limited by time slots	Higher, limited by interference
Examples	FM radio, AMPS	GSM, DECT	3G networks, GPS

145	What are the Common Multiple Access Technologies? Differentiate between Multiplexing & Multiple Access Technologies.	BUET M.Sc. 12
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Solution:

There are three types of common Multiple Access Methods :

- Frequency Division Multiple Access (FDMA) - flexible and simple
- Time Division Multiple Access (TDMA) - popular
- Code Division Multiple Access (CDMA Spread Spectrum) - highly secure



This illustration shows how the most common Multiple Access Methods allow the capacity of a standard transponder to be shared.

Difference between Multiplexing and Multiple Access:

Multiplexing	Multiple Access
Multiplexing is a process where multiple analog message signals or digital data streams are combined into one signal over a shared medium.	Multiple access method allows several terminals connected to the same multi-point physical medium to transmit over it and to share its capacity."
It works on the physical layer (L1) of OSI model.	It works on the Data Link layer (L2) of OSI model.

A device that performs the multiplexing is called a multiplexer (MUX), and a device that performs the reverse process is called a demultiplexer (DEMUX).

A channel-access scheme is also based on a Multiple access protocol and control mechanism, also known as media access control (MAC). This protocol deals with issues such as addressing, assigning multiplex channels to different users, and avoiding collisions.

158 What is meant by: OFDM, GMSK, WiMAX, DWDM, PSTN.

BUET M.Sc.
Unknown

Solution:

OFDM : Orthogonal Frequency Division Multiplexing
 GMSK : Gaussian Minimum Shift Keying
 WiMAX : Worldwide Interoperability for Microwave Access
 DWDM : Dense Wavelength Division Multiplexing
 PSTN : Public Switched Telephone Network

142 What is the carrier in Submarine Cable?

MCQ BUET
M.Sc.-13

Ans: Light